REFLEX: Reflexive Equilibrium Fixed-point Learning for Endogenous eXchanges

Analytic Stability Boundaries for Performative Market Making in OTC Corporate Bond Markets

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Abstract

In over-the-counter corporate bond markets, dealers compete for client trades by quoting bid and ask prices. Tighter quotes attract more business, but also informed customers more likely to trade ahead of adverse price moves, leaving the dealer holding the risk. As dealers increasingly use machine learning to set quotes, they retrain these models on the trades their own quotes attract, creating a feedback loop in which each model reshapes the market that generates its next training data. The question is therefore not only whether a quoting model performs well, but whether the market it creates stays stable as the model learns from it. Existing performative prediction theory gives a sharp stability condition, yet expresses it through abstract properties of the learning objective a trading desk cannot measure before deployment. We introduce Reflex, a framework that replaces those unobservable quantities with three measurable features of dealer behavior: how strongly trading volume responds to tighter quotes, how sharply the dealer's objective bends around its optimum, and how quickly informed flow increases as spreads narrow. Reflex combines these into a single retraining modulus, a pre-deployment stability margin estimated from a desk's own quote and execution history that predicts whether repeated retraining will converge or amplify itself. In simulation, predicted and measured moduli agree within 8% where the loop contracts, though the uncorrected instability threshold is optimistic (4.70 predicted, 3.17 measured). Competing dealers increase instability by 1.74× with two and 3.16× with three, against predicted 2× and 3×. Where ordinary retraining becomes unstable at modulus 1.21, a structurally anchored correction converges as blind retraining collapses. Calibrated over 36 years of public market data, stability headroom falls roughly 4.4× for investment grade and 4.3× for high yield from calm to crisis regimes.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → **Economics**; • **Theory of computation** → *Market equilibria.*

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Keywords

performative prediction, market making, OTC corporate bonds, retraining stability, systemic risk, numerical verification

1 Introduction

Corporate bonds trade over the counter. A client requests a quote, a dealer posts a half-spread h, and the dealer's inventory absorbs the flow. Quoting on these desks is increasingly algorithmic, and quoting policies are increasingly fit to historical flow.

That flow is not exogenous. A tighter quote wins more volume, and it also summons more informed, or "toxic," flow that picks the dealer off before adverse price moves. A wider quote starves both channels. The quoting policy ϕ therefore induces the flow distribution $D(\phi)$ it will next be trained on. Fitting on the induced flow, redeploying, and refitting has the structure of *repeated risk minimization* (RRM) in performative prediction [22]. There, RRM converges to a performatively stable point whenever $\varepsilon < \gamma/\beta$, where ε bounds the sensitivity of the distribution map and γ, β are the strong convexity and smoothness of the loss.

The theorem is sharp, but its constants are Lipschitz *assumptions*. A desk cannot evaluate ε, β, or γ before deploying, so it cannot know whether its own retraining cadence is stable. Similarly, a supervisor watching many dealers retrain against one shared pool of informed flow cannot know whether competition among learners is manufacturing systemic fragility. Post-trade TRACE reporting reveals the spreads that result, not the retraining loop that produced them. Both questions are quantitative, and answering either needs the constants themselves. We compute a market-level stability margin for algorithmic liquidity provision, which puts the problem in the AI-governance and systemic-risk scope of AI in finance.

To address these limitations, this paper introduces REFLEX (*Reflexive Equilibrium Fixed-point Learning for Endogenous eXchanges*), which builds that retraining loop *inside* a structural OTC market-making model in the Guéant–Lehalle–Fernández-Tapia (GLFT) family. There the loop is a scalar cobweb map, and its local counterparts of the three constants follow from microstructure primitives. We check every closed form against a learned simulator loop. We contribute six closed-form results, each tested by experiment:

R1 *An analytic stability boundary.* We give the cobweb's γ, β, and ε in closed form from fill-curve curvature, P&L scale, and toxic-flow slope, so the modulus $m = \varepsilon\beta/\gamma$ is known before deployment. Measured moduli track it within 8% where the loop contracts.

- **R2** *A performative-gradient (PerfGD) correction*, closed-form and estimated. One extra scalar per step converges past the RRM boundary, and its structurally anchored learned counterpart settles at the realized performative optimum where blind retraining collapses.
- **R3** *A multi-dealer boundary* $\varepsilon < \gamma/(N_{\text{eff}}\beta)$. Competition destabilizes the market a factor N_{eff} before any single dealer would, and a genuine shared-pool market measures 1.74× and 3.16× amplification at $N = 2, 3$.
- **R4** *Finite-sample robust certificates.* An $O(1/\sqrt{n})$ ambiguity radius bought by common random numbers (CRN) issues stable, unstable, or *undecided* verdicts, separating statistical from structural uncertainty.
- **R5** *Factor-model scaling.* The $d\times d$ modulus matrix has spectral boundary $\rho(M) < 1$ at $O(dk^2)$ cost, and $\rho(M)$ is flat from 8 to 128 bonds on our calibration.
- **R6** *Lazy deployment.* The K-step retraining map $\mu(K) = -m + c^K(1 + m)$ yields deadbeat and maximal-stability cadences in closed form. With one fitted parameter, both match the measured probe for $K \le 8$.

The system side contributes an *un-blinded* learned market-response operator whose training identifies $dD/d\phi$. It adds three complementary instruments for ε: a CRN best-response probe, an entropic optimal-transport estimate [8, 12, 23], and a fitted informed-flow curve [7]. It calibrates the simulator to 36 years of public market data [10, 13], and it builds a verification layer of 66 numerical proof certificates plus Lean 4 [9, 24] formal skeletons. Negative results and proxy-level data provenance are reported as such throughout.

2 Related Work

Perdomo et al. [22] established the convergence of RRM and its stochastic variants [19], and later work estimates the distribution response $dD/d\phi$ to optimize performative risk directly [16, 20]. The framework now reaches decision-dependent stochastic optimization [11], distributions that carry their own dynamics [5, 18], bandit feedback [17], multi-agent decision-dependent games [21], and robust formulations; see [15] for a survey. In all of it, $(\varepsilon, \beta, \gamma)$ stay abstract constants of an unspecified loss.

On the market-making side, Avellaneda and Stoikov [1] and Guéant et al. [14] derived closed-form optimal quotes under exponential fill intensities, and later work added multi-asset dimensionality reduction [4], adverse selection and price reading [3], the stochastic-control toolkit [6], and the empirical anatomy of flow–price interaction [7]. These models take the flow response to quoting as a fixed input, so the learning loop that re-estimates the market from its own induced data falls outside their scope.

To our knowledge, no prior work computes a performative stability modulus from market structure.

3 Market Model and the Retraining Loop

One deployment is one period of quoting by a dealer posting half-spread h. Skew is zero for the scalar theory, and the multi-bond lift is R5. With ρ the latent liquidity ratio, benign notional follows the GLFT exponential fill curve [1, 14] and informed notional a spread-gated channel:

$$U(h) = A\rho\, e^{-kh}, \qquad \tau(h) = \rho\,\bar{g}\big(I_b + \alpha f I\, e^{-c_t h}\big), \tag{1}$$

where A, k are the arrival rate and demand elasticity, I_b, I the base and feedback informed intensities, α adversarial informativeness, f the toxicity-feedback gain (our control variable), c_t the toxic spread decay, and $\bar{g} \in (0, 1)$ the mean of the informed traders' tanh signal gate. Informed flow trades with the signal, so each unit costs the dealer an adverse-selection severity $\psi > 0$, a state constant from the same Gaussian gate integrals. The expected one-deployment objective, with P&L scale P, quoting-cost weight w, anchor h_{ref}, and an inventory-variance penalty R_q carried through its curvature $R_q'' = \lambda_q \ge 0$, is

$$J(h; T) = P\big[\, hA\rho e^{-kh} + hT - \psi T - w(h - h_{\text{ref}})^2 - R_q(h) \,\big]. \tag{2}$$

The frozen toxic level T is what makes the learner structurally blind. Within a deployment the fitted model conditions on the deployed regime, so the dealer best-responds to $T = \tau(h_{\text{dep}})$ while only the benign channel varies with the candidate h. Retraining is therefore a cobweb map,

$$h_{t+1} = \text{BR}(h_t) := \arg\max_h J\big(h; \tau(h_t)\big), \qquad m := \big|\text{BR}'(h^*)\big|, \tag{3}$$

which contracts locally to the performatively stable spread h_{SP} if $m < 1$ and oscillates away from it if $m > 1$. The simulator realizes (1) to (3) with multi-bond flow, an informed-flow saturation cap, and a liquidity field that *inflates with realized flow*. The frozen closed forms omit those endogenous channels, and the measurements below make them visible.

4 Closed-Form Stability Theory

We state the objects the experiments test. All six follow from the one-deployment objective (2) by differentiation and the implicit-function theorem; only R1's argument is given in full.

R1 (analytic boundary). Differentiating (2) gives the cobweb's counterparts of the three performative constants in closed form. They are local values at h, not the uniform bounds of [22]:

$$\gamma = P\big[2w + A\rho k e^{-kh}(2 - kh) + \lambda_q\big], \qquad \beta = P,$$
$$\varepsilon(h) = \rho\,\bar{g}\,\alpha f I c_t\, e^{-c_t h}. \tag{4}$$

Curvature is fill-curve curvature plus quoting-cost stiffness, the coupling β is the P&L scale, and ε is the toxic-flow slope $|d\tau/dh|$, a mean shift that lower-bounds the Wasserstein sensitivity of [22]. The implicit-function theorem gives $\text{BR}' = -\varepsilon\beta/\gamma$, an oscillatory cobweb, hence

$$m(h^*) = \frac{\varepsilon\beta}{\gamma} = \frac{\rho\bar{g}\alpha f I c_t\, e^{-c_t h^*}}{2w + A\rho k\, e^{-kh^*}(2 - kh^*) + \lambda_q}, \tag{5}$$

with P cancelling and h^* the self-consistent fixed point of (3). The loop is locally stable if $m < 1$ and unstable if $m > 1$. Every term is evaluable before any loop is run, so the boundary is a *prediction*.

Derivation and regularity. Subscripts 1 and T denote partial derivatives in h and T. Where $\gamma(h) > 0$ on the capped quoting interval the frozen-T objective is C^2 and strictly concave, $\partial_{11}J = -\gamma < 0$, so its maximiser is interior and unique and $\partial_1 J(h_{t+1}; \tau(h_t)) = 0$ defines BR implicitly. Differentiating in h_t gives $\text{BR}' = -\partial_{1T}J\,\tau'/\partial_{11}J$, and (2)–(1) supply $\partial_{1T}J = \beta$ and $\tau' = -\varepsilon$, which is (5); local stability is then Banach contraction near h^*. R3, R5 and R6 reuse this step with $\partial_{1T}J$ replaced by a rank-one coupling, by $\Gamma^{-1}E$, and by the K-step warm-started map.

Two structural corollaries matter for honest measurement. First, adverse selection (ψT) is constant in h within a deployment, so it sets the profit level and the echo-chamber gap of R2 but drops out of the boundary. Second, the modulus *saturates* at the self-consistent fixed point: widening spreads decay $\varepsilon(h^*)$ exponentially, and at default-like constants it never crosses 1. This is defensive widening, so measured instability describes the retraining map at the *operating* spread, and we report it that way.

R2 (PerfGD correction). The performative objective is $\Phi(h) = J(h; \tau(h))$. Un-blinding adds one closed-form term to the blind gradient $G(h) = \partial_1 J(h; \tau(h))$:

$$\Phi'(h) = G(h) + \Delta(h), \qquad \Delta(h) = -\beta\,(h - \psi)\,\varepsilon(h), \tag{6}$$

where $\Delta = \partial_T J \cdot (d\tau/dh)$ is the distribution-response term, supplied entirely by the R1 closed forms, so the corrected ascent costs one scalar more per step than RRM [16]. Its convergence is governed not by the cobweb modulus but by the objective curvature at the performative optimum h_{PO},

$$\gamma_{\mathrm{PO}} = \gamma + \beta\varepsilon\,(2 + c_t\psi - c_t h_{\mathrm{PO}}), \tag{7}$$

which stays positive where $m > 1$, so the corrected loop converges beyond the RRM boundary. The stable-versus-optimal separation, the "echo-chamber gap" of [20], is $h_{\mathrm{SP}} - h_{\mathrm{PO}} = \beta\varepsilon(h_{\mathrm{SP}} - \psi)/(\gamma + \beta\varepsilon) = O(\varepsilon)$ in decision space and $O(\varepsilon^2)$ in value.

R3 (multi-dealer systemic risk). N dealers share one informed pool with toxic spillover $\kappa \in [0, 1]$, so flow displaced by one dealer's widening lands on the others. The joint best-response Jacobian is a rank-one common-mode coupling. Its differential eigenvalues carry $(1 - \kappa)m_1$ and its common mode $-m_1 N_{\mathrm{eff}}$ with $N_{\mathrm{eff}} = 1 + \kappa(N - 1)$, so the joint loop is stable if and only if

$$\varepsilon < \frac{\gamma}{N_{\mathrm{eff}}\,\beta}, \qquad N_c = 1/m_1 \tag{8}$$

is the critical dealer count at which a market of individually stable dealers ($m_1 < 1$) turns systemically unstable. Competition manufactures a synchronized cobweb, making fragility a market property rather than a desk property. "Systemic" here means this market-level retraining instability; we do not model losses or market failure.

R4 (robust boundary). With n probe estimates of ε (sample mean $\hat{\varepsilon}_n$, std s), the certificate

$$\hat{\varepsilon}_n + \delta_n < \frac{\gamma}{\beta}, \qquad \delta_n = \frac{z_{1-a}\,s}{\sqrt{n}}, \tag{9}$$

declares *stable*; $\hat{\varepsilon}_n - \delta_n > \gamma/\beta$ declares *unstable*; anything between is *undecided*. The parametric $O(1/\sqrt{n})$ radius is bought by the CRN pairing of the probe, since a naive finite difference of independent runs achieves only $O(n^{-1/3})$ after bias–variance balancing. Pinning a crossing needs $n_{\mathrm{req}} = O(\Delta^{-2})$ in the distance Δ to the boundary, so statistical uncertainty is separated from structural uncertainty by construction.

R5 (factor scaling). For d bonds, stability is governed by the modulus matrix $M = \beta\,\Gamma^{-1}E$ with $E = \mathrm{diag}(\varepsilon_i)$ and Γ the cross-bond curvature, and the loop contracts if and only if $\rho(M) < 1$. A k-factor covariance makes Γ diagonal-plus-low-rank, so $\rho(M)$ costs $O(dk^2)$ by Woodbury, with truncation error linear in the residual factor variance $\lambda_{k+1}(C)$. Large universes therefore stay certifiable at practical cost.

R6 (lazy deployment). Suppose each deployment takes only K inner gradient steps on the frozen-T objective (step η_h, contraction $c = 1 - \eta_h\gamma$), warm-started from the deployed spread instead of the exact best response. The outer map slope becomes

$$\mu(K) = -m + c^K(1 + m), \tag{10}$$

a one-parameter interpolation from inertia at $\mu(0) = 1$ to the exact cobweb as $\mu \to -m$. Three consequences follow in closed form. A *deadbeat* cadence $K_{\mathrm{db}} = \ln\big(m/(1 + m)\big)/\ln c$ gives $\mu = 0$ and one-shot convergence. For $m > 1$ a stability window $K \le K_{\max} = \ln\big((m - 1)/(m + 1)\big)/\ln c$ keeps an RRM-unstable market stable, the quantitative form of the greedy-versus-lazy gap in [19, 22]. And an observer who fits the plain boundary to a lazy loop reads a two-branch *effective* curvature $\gamma_{\mathrm{eff}}(K) = \gamma\, m/|\mu(K)|$, under-estimating stiffness below the equal-modulus cadence and over-estimating it above.

5 Experimental Setup

5.1 Learned Operator and the Four Loop Modes

The simulator is a multi-bond OTC market with benign and informed clients, inventory, an informed-flow saturation cap, and a latent liquidity field inflated by realized flow. The learned market-response operator T_θ (a PyTorch MLP, two tanh layers of width 64 with distribution heads over per-bond features, fit by Adam) predicts next-deployment flow distributions conditioned on a per-deployment *policy summary*, and is fit over a sliding window of past deployments. With window ≥ 2 the derivative of its prediction w.r.t. the summary, the learned $dD/d\phi$, is identified by automatic differentiation; a single-window fit is structurally blind and supplies the RRM baseline.

Four retraining modes close the loop. **(i)** Blind RRM refits on the latest induced flow. **(ii)** *PerfGD-analytic* uses the closed-form Δ of (6) as a surrogate gradient. **(iii)** *PerfGD-learned* consumes the free-form learned $dD/d\phi$ and is reported as a negative result. **(iv)** *PerfGD-structural* never asks the network for a derivative. It instead fits the theory's own response families

$$\hat{\tau}(h) = \hat{C}_0 + \hat{C}_1 e^{-\hat{c}h}, \qquad \hat{u}(h) = \hat{A}_u e^{-\hat{k}_u h}, \tag{11}$$

plus the realized severity $\hat{\psi}$, to the loop's *own* deployment history (per-bond, per-step samples over a 12-deployment window; a small quote jitter supplies the within-deployment identification the echo chamber cannot destroy). It then ascends $\hat{\Phi}' = \hat{G} + \hat{\Delta}$ with step $1/\gamma_{\hat{\mathrm{PO}}}$ under a 35% relative step cap and an anti-echo freeze that refuses to steer on unidentified fits. Its target is the *realized* performative optimum, judged against independent structural fits (Sec. 6.5). Both safeguards come from failures we measured: exponential response fits on a narrow spread window are line-degenerate and extrapolate badly, and the unconstrained early ascent rails against its spread cap.

5.2 Three Instruments for ε and the Probe Protocol

Three instruments estimate ε, and they do not measure the same object. (a) The *CRN best-response probe* runs paired deployments at $h_{\mathrm{ref}} \pm \delta$ under common random numbers. Its finite difference of best responses estimates m, and $\varepsilon = \hat{m}\gamma/\beta$ then inverts R1, so it is a

consistency check rather than an independent estimate. A signed K-step variant measures $\mu(K)$ for R6. (b) The *optimal-transport* estimate $\hat{\varepsilon} = W_1\big(D(h+\delta), D(h-\delta)\big)/2\delta$ on realized toxic flow is the Wasserstein sensitivity of [22] (Euclidean cost), which upper-bounds any mean slope. It is computed via debiased log-domain Sinkhorn divergences [8, 12] with the blur set *scale-relatively* (0.02× pooled std, tuned against the exact 1-D quantile W_1; Sec. 6.7). (c) The *fitted informed-flow curve* bins realized toxic notional against deployed spread, fits (1), and differentiates [7], a mean slope at the realized state.

A pre-run audit showed three protocol facts that matter at first order. Probe at the operating spread rather than the analytic fixed point. Keep collection jitter at 0.05, since at 0.2 the BR probe inflates ∼3×. And report medians and interquartile ranges (IQR) over 8 seeds with the R4 robust bands at every grid point. We sweep the gain f rather than adversariality α. High α drives the dealer to wide spreads where $e^{-c_t h}$ has decayed, so the α-response is non-monotone (measured hump 0.08 → 1.83 → 0.67).

5.3 Real-Data Calibration and Provenance

The simulator is calibrated per (rating × volatility regime) on a public, verified panel. It combines ∼36 years of daily and ∼70 years of monthly series (CBOE VIX as the σ proxy and regime classifier, WTI, the Fed H.15 10-year, Shiller equity data, gold and CPI) with the TRACE-derived corporate-bond factors of [10], whose liquidity risk factor is the primary ε proxy, and monthly returns for 212 real-CUSIP bonds. Half-spread proxies combine VIX-implied spreads with the illiquidity add-on of [2, 13], and $\lambda(h) = Ae^{-kh}$ intensities are fit per cell by maximum likelihood.

On provenance, plainly: this is not trade-level TRACE. Dealer-side prints, per-dealer inventories, and per-bond (A, k) would require TRACE Enhanced. Only (A, k, σ, h) are data-identified here, the toxic channel is structurally scaled by documented ratios, and the crisis-regime intensity fit is degenerate ($k = 0$, $n = 74$ days) and flagged wherever it binds. Regime *ordering* of the boundary is data-driven; absolute critical gains are not. Calibrated configurations run in per-$100-par units, and every probe width, tolerance, and OT blur is scale-relative by convention.

5.4 Verification Layer

Every load-bearing identity, inequality, and dynamical claim of R1–R6 is re-derived numerically by 66 *proof certificates*, run on both the raw and the calibrated real-unit configurations: central differences of the *composed* best-response map against the assembled $-\varepsilon\beta/\gamma$ with no shared code path, eigensolves against spectral formulas, Monte-Carlo rate fits against claimed exponents, and the dynamics run rather than assumed.

Building the layer caught real errors before the paper-grade run. It surfaced a convention mismatch in which the 1-D frozen-gradient helpers omit the inventory curvature, so their slopes are governed by $\gamma - P\lambda_q$. It also falsified two first drafts: an "always-stiffer" claim for γ_{eff}, replaced by the two-branch law of R6, and a mis-signed heavy-tail prediction for the R4 radius. The logical skeletons of all six results are additionally formalized in Lean 4 against mathlib [9, 24]. These are reviewed statements whose compilation is pending a toolchain, so the numerical certificates remain the verification of record. The full suite (152 tests, 11 experiments; Python, PyTorch, NumPy) runs CPU-only in ∼25 minutes, deterministic from (configuration, seed).

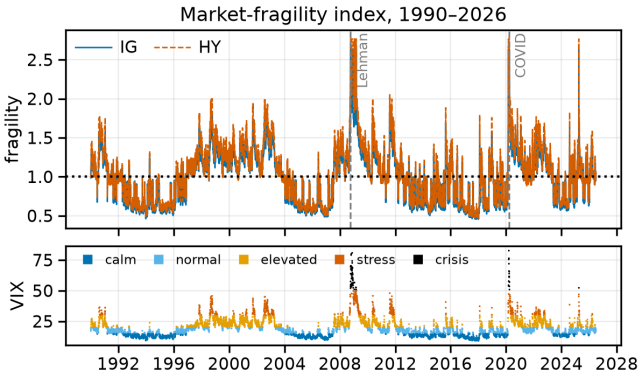


Figure 1: The daily market-fragility index, 1990–2026: the R1 closed forms evaluated on real data. Top: fragility(t) = median$(\varepsilon^*)/\varepsilon^*(t)$ for IG (solid) and HY (dashed), with the Lehman and COVID-freeze dates marked. Bottom: the VIX spine, colored by volatility regime. The crisis plateau reflects the degenerate crisis-cell fit and is flagged, not hidden.

6 Results

6.1 Numerically Certified Identities

All 66 certificates pass on the raw and the calibrated real-unit configurations. The tightest is R1's cobweb linearization, with residual 6.8×10^{-2} against a tolerance of 8×10^{-2}; the R4 rate fit sits at a third of its tolerance and every other family below 10^{-3} of its own. The R2 demo-dynamics certificate pins absolute raw-unit constants, so it runs on the raw configuration only.

6.2 A Real-Data Fragility Index (1990–2026)

Because R1 is a closed form in observables, it can be evaluated on every trading day of the panel. Figure 1 plots the resulting index, fragility(t) = median$(\varepsilon^*)/\varepsilon^*(t)$, and Table 1 gives the per-regime summary. First, the stability headroom $\varepsilon^* = \gamma/\beta$ collapses ∼4.4× for investment grade (IG) and ∼4.3× for high yield (HY) from calm to crisis. Second, HY sits more than 10× below IG in *every* regime, which is a level effect rather than a crisis effect. Third, the modulus at *observed* spreads falls into crisis (IG 0.85 → 0.14), so dealers widen faster than the toxic channel steepens. The index therefore tracks shrinking *headroom*, the smallest toxic slope that would destabilize retraining. It does not track proximity to $m = 1$, which by the same table is greatest in calm markets.

The index saturates at a crisis plateau because of the degenerate crisis fit. The GFC and the March-2020 COVID freeze are flagged at the same ceiling, so intra-crisis ranking is explicitly not identified. Its full-sample median is an ex-post normalizer, and the exercise shows that R1 can be evaluated on real data, not that it is validated there.

Table 1: Per-regime stability headroom $\varepsilon^* = \gamma/\beta$ and modulus at observed spreads, evaluated on the 1990–2026 panel.

regime	$\varepsilon^*_{\mathrm{IG}}$	$\varepsilon^*_{\mathrm{HY}}$	$m_{\mathrm{IG}}(h_{\mathrm{obs}})$	$m_{\mathrm{HY}}(h_{\mathrm{obs}})$
calm	1207.7	93.5	0.847	0.592
normal	739.2	61.3	0.870	0.563
elevated	594.4	45.9	0.520	0.367
stress	477.3	35.6	0.223	0.163
crisis	275.9	21.8	0.139	0.091

Table 2: Three-way ε triangulation at the operating spread and default gain $f = 5$, against the closed form at the a-priori and at the realized deployment state ($\rho = 2.32$, $|g| = 0.446$).

instrument	$\hat{\varepsilon}$	vs. realized
closed form, a-priori state ($\rho = 1$)	0.764	0.44×
closed form, **realized state**	1.726	1.00
CRN best-response slope	0.508	0.29×
Sinkhorn/W_1	4.578	2.65×
Fitted informed-flow curve	4.011	2.32×

6.3 Predict-Then-Verify: the Boundary Crossing

Figure 2 is the headline falsification test. We sweep the gain f (7 values × 8 seeds), overlay the a-priori curve (5), and measure the modulus with the CRN probe. In the contracting regime the agreement is quantitative: measured median 0.390 against predicted 0.426 at $f = 2$, an 8% gap. The measured crossing, interpolated between $f = 3$ and the already-unstable $f = 4$, sits at $f^* \approx 3.17$ against the a-priori 4.70, so the a-priori form over-certifies stability. Table 2 measures the deployment's own flow inflating the liquidity field to $\rho \approx 2.3$, and the closed form evaluated at that *realized* state predicts $f^* \approx 2.8$–3.0, just on the conservative side of the measurement.

The R4 certificates grade the grid exactly as the seed bands warrant: stable through $f = 2$, unstable at $f = 4$, undecided where beyond-boundary readings scatter. At $f = 8$ the eight seeds span 0.03–2.05, because past the boundary the probe is a finite-difference diagnostic rather than a local slope. The fixed-point curve itself saturates at 0.61, so the measured instability is a property of the retraining map at the operating spread, exactly as R1 predicts.

Table 2 is taken at the default gain $f = 5$, beyond the crossing, and its readings span 0.51–4.58, about 9×. The CRN slope inverts R1 from a beyond-boundary probe reading. The Sinkhorn estimate is a W_1 sensitivity and upper-bounds any mean slope. The fitted curve is a mean slope like the closed form, yet reads 2.3× above it at the realized state. That residual is the state-feedback channel a frozen-state closed form omits, so the closed form is a *lower anchor* for ε and, uncorrected, an optimistic stability certificate. A desk should evaluate R1 at the realized state.

6.4 Competition Amplifies Instability (R3)

We probe a *genuine* N-dealer market sharing one informed pool rather than an analytic shortcut; the environment reduces bit-for-bit to the single-dealer market at $N = 1$. CRN joint-mode probes measure the common-mode slope against the predicted $N_{\mathrm{eff}} m_1$

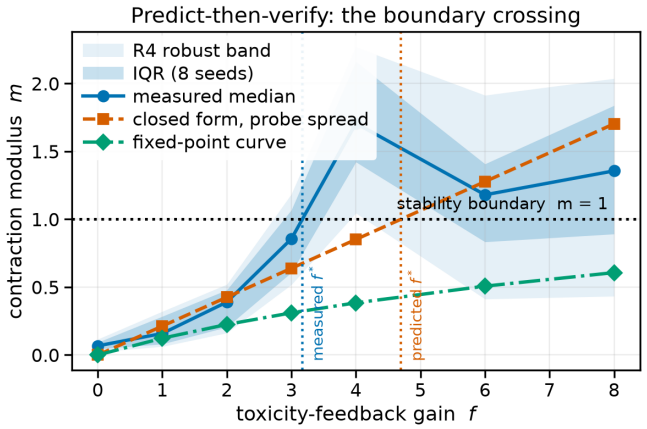


Figure 2: Predict-then-verify phase diagram over the feedback gain f: measured modulus (median, IQR band, and R4 robust band over 8 seeds) against the a-priori closed form at the probe spread and the saturating fixed-point curve, with the measured ($f^* \approx 3.17$) and predicted (4.70) crossings marked.

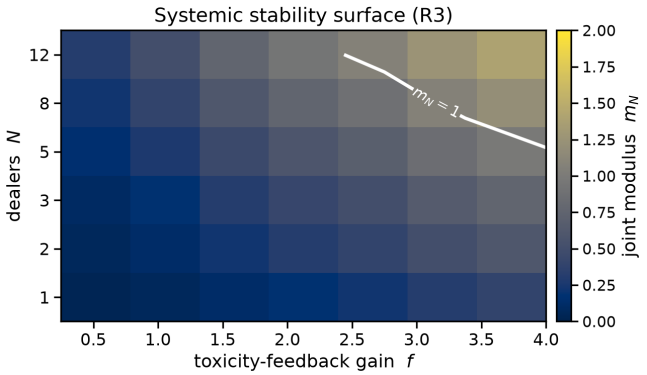


Figure 3: The R3 systemic surface: joint modulus $m_N = N_{\mathrm{eff}} m_1$ over the (f, N) grid at full spillover ($\kappa = 1$), with the $m_N = 1$ boundary contour. Gains that are comfortably stable for a single dealer cross the systemic boundary as the dealer count grows; the probes of Table 3 test the same N_{eff} scaling on the simulated market at a separate probe configuration.

(Table 3). Amplification is 1.74× at N=2 and 3.16× at N=3 against predicted 2 and 3, within 13% and 5%. The differential mode is dead at full spillover ($(1 - \kappa)m_1 = 0$), so the instability is purely the synchronized systemic channel.

The analytic surface in Figure 3 puts the critical dealer count at $N_c \approx 7.9$ for a single-dealer modulus $m_1 \approx 0.13$. The probes run at a separate interior configuration with $m_1 = 0.786$, where two dealers already cross, and the simulated three-dealer cobweb oscillates against its cap. Each desk can satisfy its own stability check while the market as a whole violates (8), so fragility has to be certified at the market level.

Table 3: Measured common-mode retraining slope on the shared-pool market versus the R3 prediction $N_{\text{eff}} m_1$ (full spillover, $\kappa = 1$; one seed; the $N = 1$ row defines m_1).

N	measured	predicted	differential mode
1	0.786	0.786	–
2	1.369	1.571	0.0034
3	2.480	2.357	0.0032

6.5 Closing the Loop-Level Gap: Structural PerfGD (R2)

The demo regime is genuinely RRM-unstable ($f = 6$, slow toxic decay, cobweb modulus 1.21). There the blind cobweb provably diverges while the corrected 1-D ascent converges, in closed form. The open question is whether a *learned* loop can realize that correction. Figure 4 and Table 4 answer it from a common seed.

Blind RRM collapses into the echo chamber, ending at half-spread 0.16. PerfGD-analytic and the free-form PerfGD-learned mode both fail to settle, because each consumes the operator's implied objective gradient. The run's seam diagnostics show why. The free-form learned toxic slope starts right-signed (−0.84 against analytic −1.79), then flips sign and hovers near zero. The network has ample capacity; what it lacks is identification off the deployed regime. The structural mode never asks it for a derivative. Fitting (11) to its own deployment history, it climbs under the step cap and *converges to the realized performative optimum* at $h = 3.193$, with late steps falling $0.35 \to 0.02$. That is within 0.7% of its own running estimate $\hat{h}_{\text{PO}} = 3.172$, and its fitted slope tracks the analytic shape (−1.33 against −2.00 at the final iterate) while strengthening as the fit window widens.

Benchmark honesty. The frozen-reference closed-form optimum ($h_{\text{PO}} = 1.641$) is *not* the target. In this high-intensity regime the realized market differs from the frozen closed forms at first order through channels they omit by construction: informed-flow saturation, the $\rho \approx 2.3$ liquidity inflation of Sec. 6.3, and severity drift. The loop is therefore verified against *independent structural fits* on fresh controlled deployments spanning the operating range. Against those fits the settle point (i) zeroes their corrected gradient (residual < 30% of its start value), (ii) sits strictly inside their blind stable point ($< 0.85 \times \hat{h}_{\text{SP}}$, the realized echo-chamber gap, closed), and (iii) brackets their independent optimum estimate. We claim only this much: un-blinding is proven in closed form, and the structurally anchored learned loop realizes it at loop level against the realized-market benchmark, while the free-form correction remains a documented negative result.

6.6 Lazy Deployment Stabilizes an Unstable Market (R6)

At a beyond-boundary configuration ($f = 5$, exact-BR anchor $\hat{m} = 1.205 > 1$ measured on the same seeds), the signed CRN K-step probe traces $\mu(K)$ for $K \in \{1, 2, 3, 5, 8, 12, 20\}$ over 3 seeds (Figure 5). A one-parameter fit of (10) gives inner contraction $c = 0.661$, and with that one fitted parameter both predicted cadences match. The deadbeat sign flip is predicted at $K_{\text{db}} = 1.46$, and measured medians run +0.021 at K=1 and −0.064 at K=2. The stability-window exit is predicted at $K_{\max} = 5.75$, and measured $|\mu|$ is 0.695 at K=5 (inside) and 1.541 at K=8 (outside).

Table 4: Loop outcomes in the RRM-unstable demo regime (10 deployments, common seed; convergence verified against independent structural fits).

mode	final h	outcome
blind RRM	0.16	echo-chamber collapse
PerfGD-analytic	0.36	no settle (operator gradient)
PerfGD-learned	0.66	no settle (negative result)
PerfGD-structural	**3.19**	**converges to realized optimum**

Laziness therefore keeps this RRM-unstable market measurably stable for $K \lesssim 6$ and inherits the cobweb's divergence beyond, which makes retraining *cadence* a stability control. The measured $\gamma_{\text{eff}}/\gamma$ falls monotonically through 1 between $K = 5$ and $K = 8$, the stiff branch of the two-branch law. The law holds only for $K \leq 8$. At $K = 12$ and 20 the medians (−4.47, −4.81) fall far below the asymptote $-m$, which (10) cannot produce: at that depth the probe compounds the beyond-boundary bifurcation. The clean tight-fit demonstration lives in the contracting regime, where measured +0.78/+0.70/+0.37 meet fitted +0.88/+0.68/+0.36 at $K = 1/3/8$.

6.7 Scaling and Estimator Tuning

Factor scaling (R5). With per-bond volatilities dispersed at the data-calibrated cross-sectional coefficient of variation, $\rho(M)$ is flat at ≈0.50 from $d = 8$ to 128 bonds (Figure 6) and pinned to the worst scalar modulus. On this calibration correlation does not manufacture cross-sectional instability, and the fragile mode is idiosyncratic. The Woodbury path computes $d = 128$ in 0.12 s and matches the dense eigensolve to machine precision; the truncation bound holds with 3–4 orders of magnitude of slack.

Sinkhorn blur (R1 instrument). Against the exact 1-D quantile W_1, the debiased-divergence bias is U-shaped in the scale-relative blur at a fixed iteration budget. The minimum sits at 0.02× std, giving 6.1% ground-truth bias and 1.2% on the config's CRN samples, and being scale-relative it transfers to real-unit configurations unchanged.

Robust radius (R4). Across normal, heavy-tailed, and skewed estimate distributions the $z\,s$ radius over-covers (0.99–1.00 at 95% nominal), and only railed-probe contamination degrades it (0.938). On the actual CRN probe estimates the quantile/$z\,s$ multiplier is 0.50, so $z\,s$ is binding.

7 Conclusion

This paper proposes REFLEX, which gives a quoting desk's retraining loop a stability condition the desk can evaluate. We compute the cobweb's $(\varepsilon, \beta, \gamma)$ from microstructure primitives, extend the boundary in closed form to correction, competition, finite samples, dimension, and cadence, test each extension predict-then-verify against a learned loop, evaluate it as a daily fragility index on 36 years of real market data, and certify the identities numerically. Two of the findings should carry past this model. Competition among retraining dealers amplifies instability by the effective dealer count, which makes certification of algorithmic liquidity provision

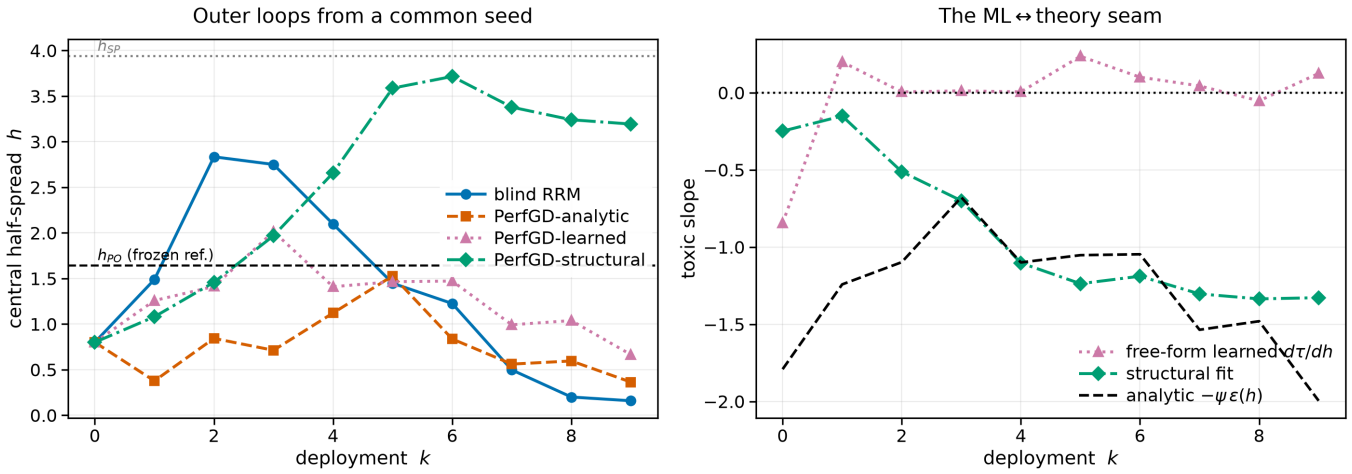


Figure 4: Left: half-spread trajectories of the four learned retraining loops from a common seed in the RRM-unstable regime (the frozen-reference h_{PO} and h_{SP} lines are context, not the benchmark; Sec. 6.5). Blind RRM collapses; the structurally anchored loop converges to the realized performative optimum. Right: the three-way ML↔theory seam, showing the free-form learned, structurally fitted, and analytic toxic slopes logged at every deployment. The free-form slope flips sign and hovers near zero; the structural fit tracks the analytic shape.

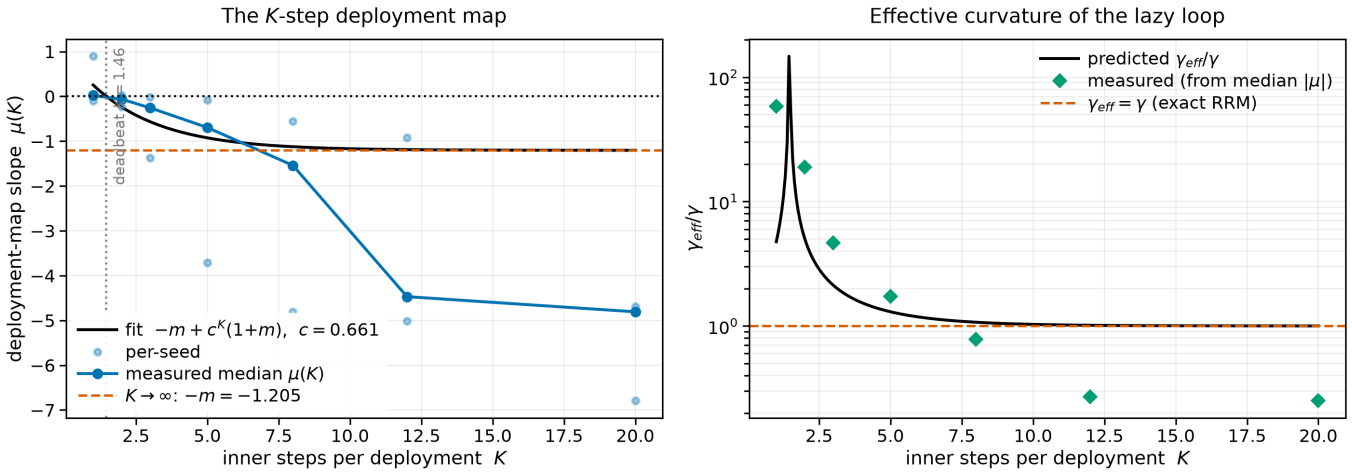


Figure 5: Lazy deployment at a beyond-boundary configuration ($\hat{m} = 1.205$). Left: the measured signed K-step map (medians over seeds; faint dots are per-seed readings) against the one-parameter fit $\mu(K) = -m + c^K(1+m)$ with $c = 0.661$, its $K \to \infty$ asymptote $-m$, and the predicted deadbeat cadence. Right: the implied effective curvature $\gamma_{\mathrm{eff}}(K)/\gamma$, predicted against measured, with the stiff branch decaying through the exact-RRM line between $K = 5$ and $K = 8$.

a market-level rather than a desk-level problem. And a learned corrected loop stabilizes where blind retraining collapses only when its response model is anchored to market structure. The free-form alternative fails with the same data and more capacity, so a desk should spend its modelling budget on structural anchoring rather than on network capacity.

These results hold under idealizing assumptions. First, the calibration is proxy-level rather than trade-level TRACE: VIX-implied spreads, no per-dealer inventories, a structurally scaled toxic channel, and a degenerate crisis cell ($k = 0$). Regime ordering is therefore data-driven while absolute critical gains are not, and TRACE Enhanced calibration removes that limit. Second, at default-like constants the self-consistent fixed point never destabilizes because dealers widen defensively, so measured crossings describe the retraining map at the operating spread. Beyond the boundary, probe readings are finite-difference diagnostics with seed-level bifurcation, so medians with robust bands carry the results, and moduli are comparable within a protocol rather than across protocols. Third, the structural loop's optimum is the *realized* one, benchmarked against independent structural fits rather than the frozen-reference closed form, with the gap attributed to saturation, liquidity inflation, and severity drift. Its convergence is to the noise ball of an estimated

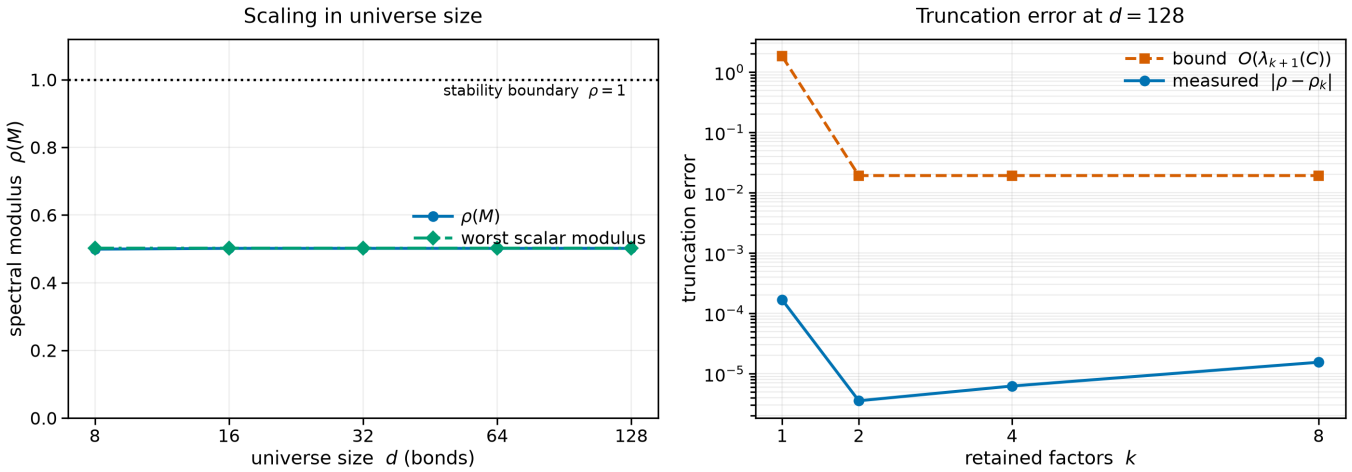


Figure 6: Factor-model scaling (R5) with data-calibrated per-bond dispersion. Left: $\rho(M)$ is flat in universe size from $d = 8$ to 128 bonds and pinned to the worst scalar modulus, far below the $\rho = 1$ boundary (dotted). Right: the truncation bound $O(\lambda_{k+1}(C))$ against the measured error $|\rho - \rho_k|$ at $d = 128$, showing three to four orders of magnitude of slack at every retained-factor count k.

gradient, and the free-form learned correction remains a negative result by design. Finally, the Lean 4 skeletons are reviewed but not yet compiled, so the 66 numerical certificates are the verification of record. Future work: trade-level calibration, inventory-state loop dynamics under state-dependent performativity, richer policy classes, and compiling the formal layer.

Reproducibility

The full framework is openly available:[1] the six derivation documents D1–D6, the simulator, the operator, all four loop modes, the estimator suite, the calibration pipeline with its data catalogue, the 66-certificate verification layer, the Lean sources, and the illustrated reports of both paper-grade runs. All results in this paper are regenerated by one command (`run_all -profile full`, ~25 CPU-minutes, deterministic from configuration and seed).

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